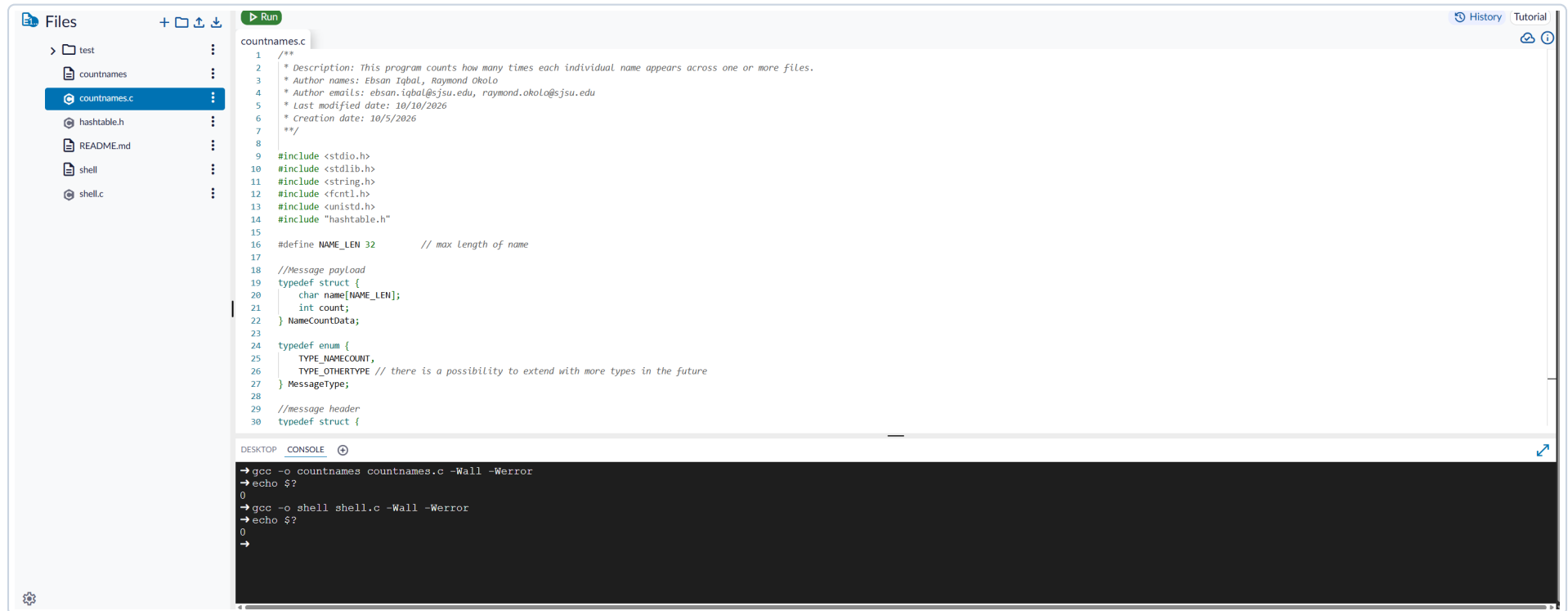


1. Clean compilation

Compiled with `gcc -o countnames countnames.c -Wall -Werror` and `gcc -o shell shell.c -Wall -Werror`; both exit status 0. `hashtable.h` is included by both programs, so no extra source file is needed.

Full screenshot: files and console

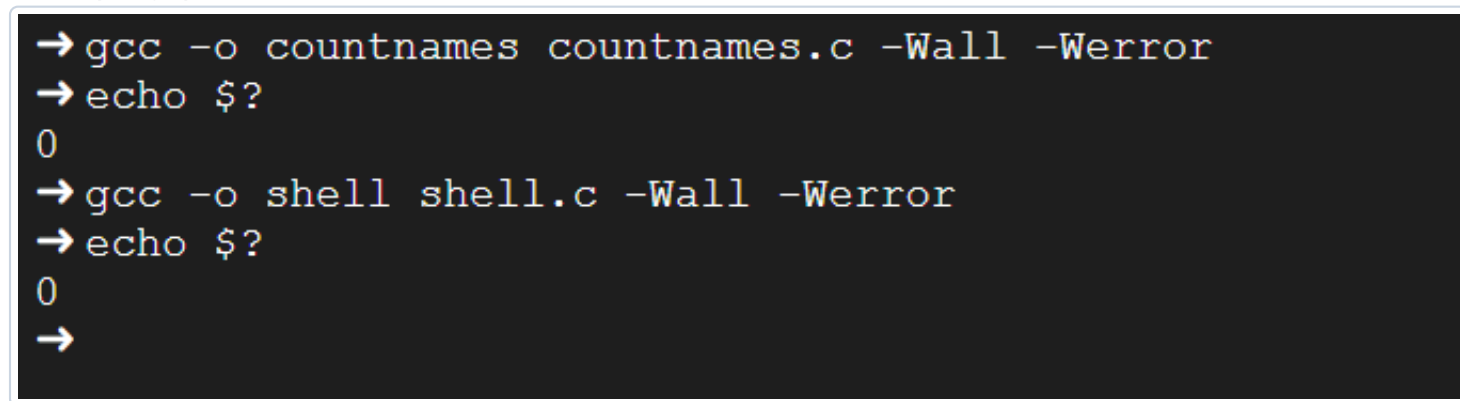


The screenshot shows a code editor with a file explorer on the left and a console at the bottom. The file explorer lists files: test, countnames, countnames.c (selected), hashtable.h, README.md, shell, and shell.c. The main editor displays the source code for `countnames.c`. The code includes a header comment, standard library includes, a macro for `NAME_LEN`, a struct for `NameCountData`, an enum for `MessageType`, and a message header struct. The console shows the compilation and execution of both programs, both resulting in exit status 0.

```
countnames.c
1 /**
2  * Description: This program counts how many times each individual name appears across one or more files.
3  * Author names: Ebsan Iqbal, Raymond Okolo
4  * Author emails: ebsan.iqbal@sjsu.edu, raymond.okolo@sjsu.edu
5  * Last modified date: 10/10/2026
6  * Creation date: 10/5/2026
7  */
8
9 #include <stdio.h>
10 #include <stdlib.h>
11 #include <string.h>
12 #include <fcntl.h>
13 #include <unistd.h>
14 #include "hashtable.h"
15
16 #define NAME_LEN 32    // max length of name
17
18 //Message payload
19 typedef struct {
20     char name[NAME_LEN];
21     int count;
22 } NameCountData;
23
24 typedef enum {
25     TYPE_NAMECOUNT,
26     TYPE_OTHERTYPE // there is a possibility to extend with more types in the future
27 } MessageType;
28
29 //message header
30 typedef struct {
```

```
DESKTOP  CONSOLE
→ gcc -o countnames countnames.c -Wall -Werror
→ echo $?
0
→ gcc -o shell shell.c -Wall -Werror
→ echo $?
0
→
```

Console (enlarged)



The enlarged console shows the same compilation and execution commands as the previous screenshot, confirming that both programs were compiled successfully and executed without errors.

```
→ gcc -o countnames countnames.c -Wall -Werror
→ echo $?
0
→ gcc -o shell shell.c -Wall -Werror
→ echo $?
0
→
```

2. Provided tests: names1.txt and names2.txt

The shell prints the totals to stdout; each child writes <PID>.out and <PID>.err.

Child 455 - test/names1.txt

Terminal

```
→ ./shell
% ./countnames test/names1.txt
Tom Wu: 3
Child 455 terminated normally with exit code: 0
% █
```

455.out (stdout)

```
countnames.c x 455.err x 455.out x
1   Tom Wu: 3
2   █
```

455.err (stderr)

```
countnames.c x 455.err x
1   Warning - file test/names1.txt line 3 is empty.
2   █
```

Child 456 - test/names2.txt

Terminal

```
% ./countnames test/names2.txt
Jenn Xu: 2
Tom Wu: 1
Child 456 terminated normally with exit code: 0
% █
```

456.out (stdout)

```
countnames.c x 456.err x 456.out x
1   Jenn Xu: 2
2   Tom Wu: 1
3   █
```

456.err (stderr)

```
countnames.c x 456.err x
1   █
```

3. Provided tests: names.txt and namesB.txt

Duplicate counting and multi-word names go to .out and the totals; empty-line warnings go to .err.

Child 469 - test/names.txt

Terminal

```
→ ./shell
% ./countnames test/names.txt
Nicky: 1
Dave Joe: 2
Yuan Cheng Chang: 3
John Smith: 1
Child 469 terminated normally with exit code: 0
% █
```

469.out (stdout)

```
countnames.c x 469.err x 469.out x █
1 Nicky: 1
2 Dave Joe: 2
3 Yuan Cheng Chang: 3
4 John Smith: 1
█
```

469.err (stderr)

```
countnames.c x 469.err x █
1 Warning - file test/names.txt line 2 is empty.
2 Warning - file test/names.txt line 5 is empty.
3 █
```

Child 470 - test/namesB.txt

Terminal

```
% ./countnames test/namesB.txt
Nicky: 1
Dave Joe: 2
Yuan Cheng Chang: 3
John Smith: 1
Child 470 terminated normally with exit code: 0
% █
```

470.out (stdout)

```
countnames.c x 469.err x 469.out x 470.err x 470.out x █
1 Nicky: 1
2 Dave Joe: 2
3 Yuan Cheng Chang: 3
4 John Smith: 1
5 █
```

470.err (stderr)

```
countnames.c x 469.err x 469.out x 470.err x █
1 Warning - file test/namesB.txt line 2 is empty.
2 Warning - file test/namesB.txt line 5 is empty.
3 █
```

4. Large provided tests (overview)

names_long.txt and the four names_long_redundant files. Each run's terminal totals, .err and .out are on the next pages.

Which child ran which file

Child	Input file	Lines in PID.out	PID.err
483	test/names_long.txt	101	empty
496	test/names_long_redundant.txt	91	line 97 is empty
509	test/names_long_redundant1.txt	40	lines 2, 4, 6, 8 are empty
522	test/names_long_redundant2.txt	44	empty
537	test/names_long_redundant3.txt	41	empty

Note: the terminal and .out screenshots on the following pages are split into a top part and a bottom part (or cut off after about 40 lines) because these outputs are longer than one screenshot. Each page says which lines are shown. Each .err file is shown in full.

4. Large test: names_long.txt (terminal)

Child 483: ./countnames test/names_long.txt - totals printed by the shell (top and bottom of the output).

Terminal: top part (left) and bottom part (right)

```
→ ./shell
% ./countnames test/names_long.txt
MARY SMITH: 1
: 99
PATRICIA JOHNSON: 1
LINDA WILLIAMS: 1
BARBARA JONES: 1
ELIZABETH BROWN: 1
JENNIFER DAVIS: 1
MARIA MILLER: 1
SUSAN WILSON: 1
MARGARET MOORE: 1
DOROTHY TAYLOR: 1
LISA ANDERSON: 1
NANCY THOMAS: 1
KAREN JACKSON: 1
BETTY WHITE: 1
HELEN HARRIS: 1
SANDRA MARTIN: 1
DONNA THOMPSON: 1
CAROL GARCIA: 1
RUTH MARTINEZ: 1
SHARON ROBINSON: 1
MICHELLE CLARK: 1
LAURA RODRIGUEZ: 1
SARAH LEWIS: 1
KIMBERLY LEE: 1
DEBORAH WALKER: 1
JESSICA HALL: 1
SHIRLEY ALLEN: 1
CYNTHIA YOUNG: 1
ANGELA HERNANDEZ: 1
MELISSA KING: 1
BRENDA WRIGHT: 1
AMY LOPEZ: 1
ANNA HILL: 1
REBECCA SCOTT: 1
VIRGINIA GREEN: 1
KATHLEEN ADAMS: 1
PAMELA BAKER: 1
MARTHA GONZALEZ: 1
DEBRA NELSON: 1
AMANDA CARTER: 1
```

```
MILDRED BAILEY: 1
KATHERINE RIVERA: 1
JOAN COOPER: 1
ASHLEY RICHARDSON: 1
JUDITH COX: 1
ROSE HOWARD: 1
JANICE WARD: 1
KELLY TORRES: 1
NICOLE PETERSON: 1
JUDY GRAY: 1
CHRISTINA RAMIREZ: 1
KATHY JAMES: 1
THERESA WATSON: 1
BEVERLY BROOKS: 1
DENISE KELLY: 1
TAMMY SANDERS: 1
IRENE PRICE: 1
JANE BENNETT: 1
LORI WOOD: 1
RACHEL BARNES: 1
MARILYN ROSS: 1
ANDREA HENDERSON: 1
KATHRYN COLEMAN: 1
LOUISE B JENKINS: 1
SARA A PERRY: 1
ANNE J POWELL: 1
JACQUELINE K LONG: 1
WANDA M PATTERSON: 1
BONNIE HUGHES: 1
JULIA FLORES: 1
RUBY WASHINGTON: 1
LOIS BUTLER: 1
TINA SIMMONS: 1
PHYLLIS FOSTER: 1
NORMA GONZALES: 1
PAULA BRYANT: 1
DIANA ALEXANDER: 1
ANNIE RUSSELL: 1
LILLIAN GRIFFIN: 1
EMILY DIAZ: 1
ROBIN HAYES: 1
Child 483 terminated normally with exit code: 0
%
```

Note: the middle of the shell's output is not shown so it fits on the page. The last line shows child 483 terminating normally with exit code 0.

483.err (stderr): empty

```
countnames.c x 483.err x
1
```

4. Large test: names_long.txt (PID.out)

Child 483: 483.out has 101 lines. Screenshots show lines 1-36 and 68-101.

483.out (stdout): top part (left) and bottom part (right)

countnames.c x 483.err x 483.out x

```
1 MARY SMITH: 1
2 : 99
3 PATRICIA JOHNSON: 1
4 LINDA WILLIAMS: 1
5 BARBARA JONES: 1
6 ELIZABETH BROWN: 1
7 JENNIFER DAVIS: 1
8 MARIA MILLER: 1
9 SUSAN WILSON: 1
10 MARGARET MOORE: 1
11 DOROTHY TAYLOR: 1
12 LISA ANDERSON: 1
13 NANCY THOMAS: 1
14 KAREN JACKSON: 1
15 BETTY WHITE: 1
16 HELEN HARRIS: 1
17 SANDRA MARTIN: 1
18 DONNA THOMPSON: 1
19 CAROL GARCIA: 1
20 RUTH MARTINEZ: 1
21 SHARON ROBINSON: 1
22 MICHELLE CLARK: 1
23 LAURA RODRIGUEZ: 1
24 SARAH LEWIS: 1
25 KIMBERLY LEE: 1
26 DEBORAH WALKER: 1
27 JESSICA HALL: 1
28 SHIRLEY ALLEN: 1
29 CYNTHIA YOUNG: 1
30 ANGELA HERNANDEZ: 1
31 MELISSA KING: 1
32 BRENDA WRIGHT: 1
33 AMY LOPEZ: 1
34 ANNA HILL: 1
35 REBECCA SCOTT: 1
36 VIRGINIA GREEN: 1
```

countnames.c x 483.err x 483.out x

```
68 KELLY TURKES: 1
69 NICOLE PETERSON: 1
70 JUDY GRAY: 1
71 CHRISTINA RAMIREZ: 1
72 KATHY JAMES: 1
73 THERESA WATSON: 1
74 BEVERLY BROOKS: 1
75 DENISE KELLY: 1
76 TAMMY SANDERS: 1
77 IRENE PRICE: 1
78 JANE BENNETT: 1
79 LORI WOOD: 1
80 RACHEL BARNES: 1
81 MARILYN ROSS: 1
82 ANDREA HENDERSON: 1
83 KATHRYN COLEMAN: 1
84 LOUISE B JENKINS: 1
85 SARA A PERRY: 1
86 ANNE J POWELL: 1
87 JACQUELINE K LONG: 1
88 WANDA M PATTERSON: 1
89 BONNIE HUGHES: 1
90 JULIA FLORES: 1
91 RUBY WASHINGTON: 1
92 LOIS BUTLER: 1
93 TINA SIMMONS: 1
94 PHYLLIS FOSTER: 1
95 NORMA GONZALES: 1
96 PAULA BRYANT: 1
97 DIANA ALEXANDER: 1
98 ANNIE RUSSELL: 1
99 LILLIAN GRIFFIN: 1
100 EMILY DIAZ: 1
101 ROBIN HAYES: 1
102
```

Note: lines 37-67 of 483.out are not visible in these screenshots; the file continues between the two images.

4. Large test: names_long_redundant.txt (terminal)

Child 496: ./countnames test/names_long_redundant.txt - repeated names are summed (Avery Taylor: 2, Andrew Lee: 2, Nathan Davis: 2).

Terminal: top part (left) and bottom part (right)

```
→ ./shell
% ./countnames test/names_long_redundant.txt
John Smith: 1
Emily Davis: 1
Michael Johnson: 1
Sophia Lee: 1
Daniel Rodriguez: 1
Mia Hernandez: 1
David Kim: 1
Olivia Brown: 1
Anthony Martinez: 1
Evelyn Jackson: 1
Christopher Gonzalez: 1
Avery Taylor: 2
Andrew Lee: 2
Aria Perez: 1
Matthew Davis: 1
Victoria Wilson: 1
William Anderson: 1
Madison Green: 1
Joshua Wright: 1
Isabella Clark: 1
Nathan Davis: 2
Natalie Adams: 1
Ethan Martin: 1
Chloe Baker: 1
Ryan Lee: 1
Samantha Perez: 1
Benjamin Wright: 1
Aaliyah Garcia: 1
Jacob Cooper: 1
Elizabeth Martinez: 1
Tyler Thompson: 1
Hannah Scott: 1
Alexander Rodriguez: 1
Makayla Turner: 1
Nicholas Nguyen: 1
Sofia Hernandez: 1
Brandon Smith: 1
Grace Lee: 1
Christian Martinez: 1
Ava Jones: 1
```

```
Luke Davis: 1
Aaliyah Brown: 1
Christopher Anderson: 1
Abigail Perez: 1
Daniel Lee: 1
Jayden Robinson: 1
Leah Lee: 1
Nicholas Jackson: 1
Avery Lee: 1
Emma Hernandez: 1
Ethan Brown: 1
Madison Martinez: 1
Lucas Davis: 1
Aria Wright: 1
Ryan Rodriguez: 1
Sophia Turner: 1
Benjamin Davis: 1
Ella Garcia: 1
Chloe Kim: 1
William Scott: 1
Ava Smith: 1
David Wright: 1
Natalie Martin: 1
Isabella Adams: 1
Brandon Davis: 1
Makayla Hernandez: 1
Alexander Clark: 1
Emily Rodriguez: 1
Christopher Lee: 1
Olivia Lee: 1
Samantha Martinez: 1
Daniel Anderson: 1
Mia Lee: 1
Jacob Wright: 1
Elizabeth Adams: 1
Tyler Martinez: 1
Hannah Garcia: 1
Nicholas Davis: 1
Sofia Thompson: 1
Brandon Brown: 1
Child 496 terminated normally with exit code: 0
% █
```

Note: the middle of the shell's output is not shown so it fits on the page. The last line shows child 496 terminating normally with exit code 0.

496.err (stderr)

```
countnames.c x 496.err x
1 Warning - file test/names_long_redundant.txt line 97 is empty.
2
```

4. Large test: names_long_redundant.txt (PID.out)

Child 496: 496.out has 91 lines. Screenshots show lines 1-39 and 54-91.

496.out (stdout): top part (left) and bottom part (right)

countnames.c x 496.err x 496.out x

```
1 John Smith: 1
2 Emily Davis: 1
3 Michael Johnson: 1
4 Sophia Lee: 1
5 Daniel Rodriguez: 1
6 Mia Hernandez: 1
7 David Kim: 1
8 Olivia Brown: 1
9 Anthony Martinez: 1
10 Evelyn Jackson: 1
11 Christopher Gonzalez: 1
12 Avery Taylor: 2
13 Andrew Lee: 2
14 Aria Perez: 1
15 Matthew Davis: 1
16 Victoria Wilson: 1
17 William Anderson: 1
18 Madison Green: 1
19 Joshua Wright: 1
20 Isabella Clark: 1
21 Nathan Davis: 2
22 Natalie Adams: 1
23 Ethan Martin: 1
24 Chloe Baker: 1
25 Ryan Lee: 1
26 Samantha Perez: 1
27 Benjamin Wright: 1
28 Aaliyah Garcia: 1
29 Jacob Cooper: 1
30 Elizabeth Martinez: 1
31 Tyler Thompson: 1
32 Hannah Scott: 1
33 Alexander Rodriguez: 1
34 Makayla Turner: 1
35 Nicholas Nguyen: 1
36 Sofia Hernandez: 1
37 Brandon Smith: 1
38 Grace Lee: 1
39 Christian Martinez: 1
```

countnames.c x 496.err x 496.out x

```
54 Christopher Anderson: 1
55 Abigail Perez: 1
56 Daniel Lee: 1
57 Jayden Robinson: 1
58 Leah Lee: 1
59 Nicholas Jackson: 1
60 Avery Lee: 1
61 Emma Hernandez: 1
62 Ethan Brown: 1
63 Madison Martinez: 1
64 Lucas Davis: 1
65 Aria Wright: 1
66 Ryan Rodriguez: 1
67 Sophia Turner: 1
68 Benjamin Davis: 1
69 Ella Garcia: 1
70 Chloe Kim: 1
71 William Scott: 1
72 Ava Smith: 1
73 David Wright: 1
74 Natalie Martin: 1
75 Isabella Adams: 1
76 Brandon Davis: 1
77 Makayla Hernandez: 1
78 Alexander Clark: 1
79 Emily Rodriguez: 1
80 Christopher Lee: 1
81 Olivia Lee: 1
82 Samantha Martinez: 1
83 Daniel Anderson: 1
84 Mia Lee: 1
85 Jacob Wright: 1
86 Elizabeth Adams: 1
87 Tyler Martinez: 1
88 Hannah Garcia: 1
89 Nicholas Davis: 1
90 Sofia Thompson: 1
91 Brandon Brown: 1
92
```

Note: lines 40-53 of 496.out are not visible in these screenshots; the file continues between the two images.

4. Large test: names_long_redundant1.txt

Child 509: ./countnames test/names_long_redundant1.txt - four empty lines (2, 4, 6, 8) are warned about in .err and ignored for counting.

Terminal (all 40 totals)

```
% ./countnames test/names_long_redundant1.txt
MARY SMITH: 1
PATRICIA JOHNSON: 1
LINDA WILLIAMS: 1
BARBARA JONES: 1
ELIZABETH BROWN: 1
JENNIFER DAVIS: 1
MARIA MILLER: 1
SUSAN WILSON: 1
MARGARET MOORE: 1
DOROTHY TAYLOR: 1
LISA ANDERSON: 1
NANCY THOMAS: 1
KAREN JACKSON: 1
BETTY WHITE: 1
HELEN HARRIS: 1
SANDRA MARTIN: 1
DONNA THOMPSON: 1
CAROL GARCIA: 1
RUTH MARTINEZ: 1
SHARON ROBINSON: 1
MICHELLE CLARK: 1
LAURA RODRIGUEZ: 1
SARAH LEWIS: 1
KIMBERLY LEE: 1
DEBORAH WALKER: 1
JESSICA HALL: 1
SHIRLEY ALLEN: 1
CYNTHIA YOUNG: 1
ANGELA HERNANDEZ: 1
MELISSA KING: 1
BRENDA WRIGHT: 1
AMY LOPEZ: 1
ANNA HILL: 1
REBECCA SCOTT: 1
VIRGINIA GREEN: 1
KATHLEEN ADAMS: 1
PAMELA BAKER: 1
MARTHA GONZALEZ: 1
DEBRA NELSON: 1
AMANDA CARTER: 1
Child 509 terminated normally with exit code: 0
```

509.out (stdout): all 40 lines

```
countnames.c x 509.err x 509.out x
1 MARY SMITH: 1
2 PATRICIA JOHNSON: 1
3 LINDA WILLIAMS: 1
4 BARBARA JONES: 1
5 ELIZABETH BROWN: 1
6 JENNIFER DAVIS: 1
7 MARIA MILLER: 1
8 SUSAN WILSON: 1
9 MARGARET MOORE: 1
10 DOROTHY TAYLOR: 1
11 LISA ANDERSON: 1
12 NANCY THOMAS: 1
13 KAREN JACKSON: 1
14 BETTY WHITE: 1
15 HELEN HARRIS: 1
16 SANDRA MARTIN: 1
17 DONNA THOMPSON: 1
18 CAROL GARCIA: 1
19 RUTH MARTINEZ: 1
20 SHARON ROBINSON: 1
21 MICHELLE CLARK: 1
22 LAURA RODRIGUEZ: 1
23 SARAH LEWIS: 1
24 KIMBERLY LEE: 1
25 DEBORAH WALKER: 1
26 JESSICA HALL: 1
27 SHIRLEY ALLEN: 1
28 CYNTHIA YOUNG: 1
29 ANGELA HERNANDEZ: 1
30 MELISSA KING: 1
31 BRENDA WRIGHT: 1
32 AMY LOPEZ: 1
33 ANNA HILL: 1
34 REBECCA SCOTT: 1
35 VIRGINIA GREEN: 1
36 KATHLEEN ADAMS: 1
37 PAMELA BAKER: 1
38 MARTHA GONZALEZ: 1
39 DEBRA NELSON: 1
40 AMANDA CARTER: 1
```

509.err (stderr)

```
countnames.c x 509.err x
1 Warning - file test/names_long_redundant1.txt line 2 is empty.
2 Warning - file test/names_long_redundant1.txt line 4 is empty.
3 Warning - file test/names_long_redundant1.txt line 6 is empty.
4 Warning - file test/names_long_redundant1.txt line 8 is empty.
5
```

4. Large test: names_long_redundant2.txt (terminal)

Child 522: ./countnames test/names_long_redundant2.txt - no empty lines, so 522.err is empty.

Terminal: top part (left) and bottom part (right), continuing without a gap

```
% ./countnames test/names_long_redundant2.txt
STEPHANIE MITCHELL: 1
CAROLYN PEREZ: 1
CHRISTINE ROBERTS: 1
MARIE TURNER: 1
JANET PHILLIPS: 1
CATHERINE CAMPBELL: 1
FRANCES PARKER: 1
ANN EVANS: 1
JOYCE EDWARDS: 1
DIANE COLLINS: 1
ALICE STEWART: 1
JULIE SANCHEZ: 1
HEATHER MORRIS: 1
TERESA ROGERS: 1
DORIS REED: 1
GLORIA COOK: 1
EVELYN MORGAN: 1
JEAN BELL: 1
CHERYL MURPHY: 1
MILDRED BAILEY: 1
KATHERINE RIVERA: 1
JOAN COOPER: 1
ASHLEY RICHARDSON: 1
JUDITH COX: 1
ROSE HOWARD: 1
JANICE WARD: 1
KELLY TORRES: 1
NICOLE PETERSON: 1
JUDY GRAY: 1
CHRISTINA RAMIREZ: 1
KATHY JAMES: 1
THERESA WATSON: 1
BEVERLY BROOKS: 1
```

```
DENISE KELLY: 1
TAMMY SANDERS: 1
IRENE PRICE: 1
JANE BENNETT: 1
LORI WOOD: 1
RACHEL BARNES: 1
MARILYN ROSS: 1
ANDREA HENDERSON: 1
KATHRYN COLEMAN: 1
LOUISE B JENKINS: 1
SARA A PERRY: 1
Child 522 terminated normally with exit code: 0
% █
```

522.err (stderr): empty

```
countnames.c x 522.err x
1
```

4. Large test: names_long_redundant2.txt (PID.out)

Child 522: 522.out has 44 lines. The screenshot shows lines 1-40.

522.out (stdout): lines 1-40

countnames.c x 522.err x 522.out x

```
1  STEPHANIE MITCHELL: 1
2  CAROLYN PEREZ: 1
3  CHRISTINE ROBERTS: 1
4  MARIE TURNER: 1
5  JANET PHILLIPS: 1
6  CATHERINE CAMPBELL: 1
7  FRANCES PARKER: 1
8  ANN EVANS: 1
9  JOYCE EDWARDS: 1
10 DIANE COLLINS: 1
11 ALICE STEWART: 1
12 JULIE SANCHEZ: 1
13 HEATHER MORRIS: 1
14 TERESA ROGERS: 1
15 DORIS REED: 1
16 GLORIA COOK: 1
17 EVELYN MORGAN: 1
18 JEAN BELL: 1
19 CHERYL MURPHY: 1
20 MILDRED BAILEY: 1
21 KATHERINE RIVERA: 1
22 JOAN COOPER: 1
23 ASHLEY RICHARDSON: 1
24 JUDITH COX: 1
25 ROSE HOWARD: 1
26 JANICE WARD: 1
27 KELLY TORRES: 1
28 NICOLE PETERSON: 1
29 JUDY GRAY: 1
30 CHRISTINA RAMIREZ: 1
31 KATHY JAMES: 1
32 THERESA WATSON: 1
33 BEVERLY BROOKS: 1
34 DENISE KELLY: 1
35 TAMMY SANDERS: 1
36 IRENE PRICE: 1
37 JANE BENNETT: 1
38 LORI WOOD: 1
39 RACHEL BARNES: 1
40 MARILYN ROSS: 1
```

Note: lines 41-44 of 522.out are not visible in this screenshot; the terminal screenshots on the previous page show those names.

4. Large test: names_long_redundant3.txt

Child 537: ./countnames test/names_long_redundant3.txt - several names appear twice (MARY SMITH: 2, PATRICIA JOHNSON: 2, ...).

Terminal (all 41 totals)

```
% ./countnames test/names_long_redundant3.txt
ANNE J POWELL: 1
JACQUELINE K LONG: 1
WANDA M PATTERSON: 1
BONNIE HUGHES: 1
JULIA FLORES: 1
RUBY WASHINGTON: 1
LOIS BUTLER: 1
TINA SIMMONS: 1
PHYLLIS FOSTER: 1
NORMA GONZALES: 1
PAULA BRYANT: 1
DIANA ALEXANDER: 1
ANNIE RUSSELL: 1
LILLIAN GRIFFIN: 1
EMILY DIAZ: 1
ROBIN HAYES: 1
MARY SMITH: 2
PATRICIA JOHNSON: 2
LINDA WILLIAMS: 2
BARBARA JONES: 2
ELIZABETH BROWN: 2
JENNIFER DAVIS: 1
MARIA MILLER: 1
SUSAN WILSON: 1
MARGARET MOORE: 1
DOROTHY TAYLOR: 1
LISA ANDERSON: 1
NANCY THOMAS: 1
KAREN JACKSON: 1
BETTY WHITE: 1
HELEN HARRIS: 1
SANDRA MARTIN: 1
DONNA THOMPSON: 1
CAROL GARCIA: 1
RUTH MARTINEZ: 1
SHARON ROBINSON: 1
MICHELLE CLARK: 1
LAURA RODRIGUEZ: 1
SARAH LEWIS: 1
KIMBERLY LEE: 1
DEBORAH WALKER: 1
Child 537 terminated normally with exit code: 0
```

537.out (stdout): lines 1-40 of 41

```
countnames.c x 537.err x 537.out x
1 ANNE J POWELL: 1
2 JACQUELINE K LONG: 1
3 WANDA M PATTERSON: 1
4 BONNIE HUGHES: 1
5 JULIA FLORES: 1
6 RUBY WASHINGTON: 1
7 LOIS BUTLER: 1
8 TINA SIMMONS: 1
9 PHYLLIS FOSTER: 1
10 NORMA GONZALES: 1
11 PAULA BRYANT: 1
12 DIANA ALEXANDER: 1
13 ANNIE RUSSELL: 1
14 LILLIAN GRIFFIN: 1
15 EMILY DIAZ: 1
16 ROBIN HAYES: 1
17 MARY SMITH: 2
18 PATRICIA JOHNSON: 2
19 LINDA WILLIAMS: 2
20 BARBARA JONES: 2
21 ELIZABETH BROWN: 2
22 JENNIFER DAVIS: 1
23 MARIA MILLER: 1
24 SUSAN WILSON: 1
25 MARGARET MOORE: 1
26 DOROTHY TAYLOR: 1
27 LISA ANDERSON: 1
28 NANCY THOMAS: 1
29 KAREN JACKSON: 1
30 BETTY WHITE: 1
31 HELEN HARRIS: 1
32 SANDRA MARTIN: 1
33 DONNA THOMPSON: 1
34 CAROL GARCIA: 1
35 RUTH MARTINEZ: 1
36 SHARON ROBINSON: 1
37 MICHELLE CLARK: 1
38 LAURA RODRIGUEZ: 1
39 SARAH LEWIS: 1
40 KIMBERLY LEE: 1
```

537.err (stderr): empty

```
countnames.c x 537.err x
1
```

Note: line 41 of 537.out (DEBORAH WALKER: 1) is not visible in the .out screenshot, but it is the last line in the terminal screenshot.

5. Multiple files: names1.txt and names2.txt

One command with two files: the shell forks two children, adds their counts (Tom Wu: 3 + 1 = 4) and prints the merged totals.

Terminal

```
→ ./shell
% ./countnames test/names1.txt test/names2.txt
Tom Wu: 4
Jenn Xu: 2
Child 550 terminated normally with exit code: 0
Child 551 terminated normally with exit code: 0
% 
```

Child 550 - test/names1.txt

550.out (stdout)

```
countnames.c x 550.err x 550.out x
1   Tom Wu: 3
2
```

550.err (stderr)

```
countnames.c x 550.err x
1   Warning - file test/names1.txt line 3 is empty.
2
```

Child 551 - test/names2.txt

551.out (stdout)

```
countnames.c x 550.err x 550.out x 551.err x 551.out x
1   Jenn Xu: 2
2   Tom Wu: 1
3
```

551.err (stderr)

```
countnames.c x 550.err x 550.out x 551.err x
1
```

6. Multiple files: three files

Three children, three pipes. Child lines return as 565, 564, 566 (not the order the files were given), which shows the children run in parallel.

Terminal

```
→ ./shell
% ./countnames test/names1.txt test/names2.txt test/mytestA2.txt
Tom Wu: 4
Jenn Xu: 2
Alice: 3
Bob: 2
Charlie: 1
Child 565 terminated normally with exit code: 0
Child 564 terminated normally with exit code: 0
Child 566 terminated normally with exit code: 0
% █
```

Child 564 - test/names1.txt

564.out (stdout) on the left, 564.err (stderr) on the right

```
countnames.c x 564.err x 564.out x
1   Tom Wu: 3
2   █
```

```
countnames.c x 564.err x
1   Warning - file test/names1.txt line 3 is empty.
2   █
```

Child 565 - test/names2.txt

565.out (stdout) on the left, 565.err (stderr) on the right

```
countnames.c x 564.err x 564.out x 565.err x 565.out x
1   Jenn Xu: 2
2   Tom Wu: 1
3   █
```

```
countnames.c x 564.err x 564.out x 565.err x
1   █
```

Child 566 - test/mytestA2.txt

566.out (stdout) on the left, 566.err (stderr) on the right

```
countnames.c x 564.err x 564.out x 565.err x 565.out x 566.err x 566.out x
1   Alice: 3
2   Bob: 2
3   Charlie: 1
4   █
```

```
countnames.c x 564.err x 564.out x 565.err x 565.out x 566.err x
1   █
```

7. Same file twice: names1.txt

The same file passed twice creates two independent children; the shell sums their counts (Tom Wu: 3 + 3 = 6).

Terminal

```
→ ./shell
% ./countnames test/names1.txt test/names1.txt
Tom Wu: 6
Child 580 terminated normally with exit code: 0
Child 579 terminated normally with exit code: 0
%
```

Child 579 - test/names1.txt

579.out (stdout)

```
countnames.c x 579.err x 579.out x
1 Tom Wu: 3
2
```

579.err (stderr)

```
countnames.c x 579.err x
1 Warning - file test/names1.txt line 3 is empty.
2
```

Child 580 - test/names1.txt

580.out (stdout)

```
countnames.c x 579.err x 579.out x 580.err x 580.out x
1 Tom Wu: 3
2
```

580.err (stderr)

```
countnames.c x 579.err x 579.out x 580.err x
1 Warning - file test/names1.txt line 3 is empty.
2
```

8. Student test: mytestA2.txt twice

Our own test file passed twice; each child counts Alice: 3, Bob: 2, Charlie: 1 and the shell prints Alice: 6, Bob: 4, Charlie: 2.

Terminal

```
→ ./shell
% ./countnames test/mytestA2.txt test/mytestA2.txt
Alice: 6
Bob: 4
Charlie: 2
Child 593 terminated normally with exit code: 0
Child 594 terminated normally with exit code: 0
%
```

Child 593 - test/mytestA2.txt

593.out (stdout)

```
countnames.c x 593.err x 593.out x
1 Alice: 3
2 Bob: 2
3 Charlie: 1
4
```

593.err (stderr)

```
countnames.c x 593.err x
1
```

Child 594 - test/mytestA2.txt

594.out (stdout)

```
countnames.c x 593.err x 593.out x 594.err x 594.out x
1 Alice: 3
2 Bob: 2
3 Charlie: 1
4
```

594.err (stderr)

```
countnames.c x 593.err x 593.out x 594.err x
1
```

9. Student tests: namesA.txt and namesC.txt

namesA.txt has two empty lines in a row (lines 4 and 5); namesC.txt has an empty first line and empty lines 4 and 7.

Child 607 - test/namesA.txt

Terminal

```
→ ./shell
% ./countnames test/namesA.txt
Tom Holland: 1
Zendaya: 1
Hank Hill: 1
Oprah: 1
Winston: 1
Child 607 terminated normally with exit code: 0
% 
```

607.out (stdout)

```
countnames.c x 607.err x 607.out x
1 Tom Holland: 1
2 Zendaya: 1
3 Hank Hill: 1
4 Oprah: 1
5 Winston: 1
6
```

607.err (stderr)

```
countnames.c x 607.err x
1 Warning - file test/namesA.txt line 4 is empty.
2 Warning - file test/namesA.txt line 5 is empty.
3
```

Child 620 - test/namesC.txt

Terminal

```
→ ./shell
% ./countnames test/namesC.txt
Mark Ruffalo: 1
Christian Bale: 1
Obama: 1
Trump: 1
Hilary: 1
Child 620 terminated normally with exit code: 0
% 
```

620.out (stdout)

```
countnames.c x 607.err x 607.out x 620.err x 620.out x
1 Mark Ruffalo: 1
2 Christian Bale: 1
3 Obama: 1
4 Trump: 1
5 Hilary: 1
6
```

620.err (stderr)

```
countnames.c x 607.err x 607.out x 620.err x
1 Warning - file test/namesC.txt line 1 is empty.
2 Warning - file test/namesC.txt line 4 is empty.
3 Warning - file test/namesC.txt line 7 is empty.
4
```

10. Totals reset between commands

Two commands in the same shell session: the totals from the first command do not carry over into the second one.

Terminal

```
→ ./shell
% ./countnames test/names1.txt
Tom Wu: 3
Child 553 terminated normally with exit code: 0
% ./countnames test/names2.txt
Jenn Xu: 2
Tom Wu: 1
Child 554 terminated normally with exit code: 0
%
```

Child 553 - test/names1.txt (first command)

553.out (stdout)

```
countnames.c x 553.err x 553.out x
1   Tom Wu: 3
2
```

553.err (stderr)

```
countnames.c x 553.err x
1   Warning - file test/names1.txt line 3 is empty.
2
```

Child 554 - test/names2.txt (second command)

554.out (stdout)

```
countnames.c x 553.err x 553.out x 554.err x 554.out x
1   Jenn Xu: 2
2   Tom Wu: 1
3
```

554.err (stderr)

```
countnames.c x 553.err x 553.out x 554.err x
1
```

Note: the first command prints Tom Wu: 3. The second command prints only Jenn Xu: 2 and Tom Wu: 1, not Tom Wu: 4, so the totals were reset between the two commands. 553.err has the empty-line warning for names1.txt line 3; 554.err is empty.

11. Standard input

`./countnames` run with no file arguments (from the shell) reads from standard input. Typed lines are counted until end of input; the empty line 4 is warned about in `stderr` and ignored.

Child 633 - standard input (no file)

Terminal

```
→ ./shell
% ./countnames
apple pineapple pen
will this work
will this work

or not?
apple pineapple pen: 1
will this work: 2
or not?: 1
Child 633 terminated normally with exit code: 0
%
```

633.out (stdout)

```
countnames.c x 633.err x 633.out x
1  apple pineapple pen: 1
2  will this work: 2
3  or not?: 1
4
```

633.err (stderr)

```
countnames.c x 633.err x
1  Warning - file stdin line 4 is empty.
2
```

Note: the input typed was: apple pineapple pen / will this work / will this work / (empty line) / or not?. Output: apple pineapple pen: 1, will this work: 2, or not?: 1. The warning in 633.err names the file as `stdin`.

12. Piping a file directly into countnames

Run from a regular bash prompt (not inside ./shell): `cat test/names2.txt | ./countnames` reads the file from standard input.

Terminal (bash)

```
→ cat test/names2.txt | ./countnames
→
```

516.out (stdout)

```
countnames.c x 516.err x 516.out x
1 Jenn Xu: 2
2 Tom Wu: 1
3
```

516.err (stderr): empty

```
countnames.c x 516.err x
1
```

Note: nothing is printed in the terminal because the totals are printed by the shell, not by countnames. countnames itself writes 516.out (Jenn Xu: 2, Tom Wu: 1) and an empty 516.err. Our shell does not support the `|` operator, so the pipe was run from bash.

13. Invalid file

`./countnames` run on a file that does not exist: `test/file_doesnt_exist.txt`.

Terminal

```
→ ./shell
% ./countnames test/file_doesnt_exist.txt
error: cannot open file test/file_doesnt_exist.txt
Child 529 terminated normally with exit code: 1
% 
```

Note: the error message is shown in the terminal and child 529 terminates with exit code 1, so the failure is reported instead of being treated as a normal run.

14. Exit

Typing `exit` at the shell prompt ends the shell and returns to the regular prompt.

Terminal

```
→ ./shell
% exit
→
```